Kitchen Manager — Project Architecture & Status Report

Executive Summary

Kitchen Manager is an end-to-end artisanal restaurant operating system and Point of Sale (POS) platform. It provides a dual-facing experience: a customer-facing digital menu with table booking, cart, checkout, UPI gateway, and live tracking; and an operations suite encompassing Kitchen Display System (KDS), Cashier Billing POS, Table Management, and Manager Analytics.

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1. Technology Stack & Architecture

Layer	Technologies Used	Description
Frontend Framework	React 19, TypeScript 5.8, Vite 6	Fast single-page application with type safety
Styling & Animation	Tailwind CSS v4, Motion (Framer Motion), Lucide Icons	Responsive modern styling, fluid transitions, and iconography
Backend & API	Node.js, Express 4.21, tsx	REST endpoints for orders, tables, dishes, settings, and auth
Data & Persistence	Dual Database Engine: SQLite/In-memory fallback + Supabase & Firebase Firestore integration	Hybrid storage supporting offline-ready operations, cloud sync, and persistent settings
Visual Analytics	Recharts 3.10	Revenue trends, top-selling dishes, order statuses, and peak hours

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2. Core Functional Modules

A. Customer Dining & Ordering Portal (/components/bistro/)

- **Interactive Menu & Categories**: Browse appetizers, mains, beverages, and chef specials with spice level indicators, dietary tags (Veg/Non-Veg/Vegan), and high-resolution food imagery.
- **Table Reservation Flow**: Modal-driven table booking with guest counts, date, time slot, and special requests.
- **Smart Cart & Customization**: Dynamic cart drawer with quantity adjustment, note additions, and GST breakdown.
- **UPI Payment Gateway Simulation**: Embedded checkout supporting UPI intent/QR flows, transaction ID generation, and immediate order verification.
- **Live Order Tracking**: Stage-by-stage status tracker (*Placed* ➔ *Preparing* ➔ *Served* ➔ *Completed*).
- **Order History & Profile**: Customer order history with bill re-printing and status inspection.

B. Operations & Kitchen Display System (KDS)

- **Live Kitchen Queue**: Real-time incoming orders with elapsed timers, urgent alerts, dish preparation checklists, and state progression (*Mark In-Prep*, *Mark Ready*).
- **Kitchen History Modal**: Comprehensive logs of completed kitchen tickets for audit and performance metrics.

C. Cashier & POS Terminal

- **Counter Order Creation**: Quick walk-in and table-order entry with dynamic dish search and item modifiers.
- **Bill Generation & GST Invoicing**: Calculates item subtotal, applicable GST (5%), discount codes, and prints branded thermal/A4 tax invoices.
- **Payment Settlement**: Settles pending "Pay at Counter" tickets and verifies digital receipts.

D. Table & Floor Management

- **Interactive Floor Plan**: Grid/card layout showing real-time occupancy (*Available*, *Occupied*, *Reserved*, *Billed*).
- **Table Operations**: Assign orders to tables, mark tables as cleared/sanitized, and handle reservations.

E. Management & Administrative Controls

- **Manager Dashboard & Analytics**: High-level KPI widgets (Gross Revenue, Active Tables, Average Order Value, Pending Orders) and Recharts graphs.
- **Menu Engineering**: Modal for adding, editing, pausing, and deleting dishes with pricing and categorization.
- **Operating Policy (Prepaid Dining Toggle)**:
    - **Prepaid Mode Enabled**: Enforces pre-payment via UPI before orders are dispatched to the kitchen. Disables "Pay at Counter".
    - **Prepaid Mode Disabled (Post-paid)**: Allows traditional dine-in where guests pay after their meal at the counter.

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3. API Endpoints Overview

Method	Endpoint	Functionality
GET / PATCH	/api/settings	Retrieve or update store settings (restaurant name, GST rate, prepaid_only flag)
GET / POST	/api/orders	Fetch live orders or submit new customer/cashier orders
PATCH	/api/orders/:id	Update order status (PREPARING, SERVED, COMPLETED, PAID)
GET / POST / PUT / DELETE	/api/dishes	Full CRUD operations for menu catalog items
GET / PATCH	/api/tables	View floor state, seat occupancy, and update table statuses
POST	/api/auth/login	Staff / Manager authentication and role verification

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4. Recent Stability & Bugfix Enhancements

- **Settings Normalization**: Resolved state desynchronization between backend JSON structure and React client, ensuring prepaid_only reliably toggles without reverting.

- **Optimistic UI & Storage Synchronization**: Instant UI button feedback backed by localStorage and cross-tab storage sync.
- **Checkout Security Rule**: Dynamic locking of payment methods preventing unpaid order submissions when prepaid enforcement is active.

5. Build & Health Status

- **TypeScript Linting**: Passed (tsc --noEmit clean, 0 errors).
- **Production Build**: Compiles cleanly with Vite + ESBuild bundle (dist/server.cjs).
- **Dev Server**: Running healthy on port 3000.

lightbulb_tipsSuggestions